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# EAS vs. Friston Free Energy Principle: A Deep Comparative Analysis

Jing Zhang

Independent Researcher

ORCID: 0009-0008-3136-2457

July 2, 2026

## Abstract

The Epistemic Axiomatic System (EAS) and Karl Friston’s Free Energy Principle (FEP) represent two ambitious frameworks for understanding intelligence and cognition. This paper provides a systematic comparison, establishing that the two frameworks are “orthogonal but in dialogue”: FEP describes how cognitive systems operate (minimize free energy), while EAS delineates what cognitive systems cannot achieve (perfect self-modeling is structurally impossible). We identify five core divergences, trace the latest Friston literature (2024–2026), and propose that EAS’s single axiom can provide first-principles grounding for FEP’s claim that approximate inference is necessarily approximate.

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# 1 Summary

EAS’s single axiom  $S \subset E \implies R_S \not\cong E$  (finite observation is necessarily a quotient of reality; perfect self-modeling is impossible) and Friston’s variational free energy minimization stand in a relationship that is **neither purely complementary nor purely competitive, but “orthogonal but in dialogue.”** FEP proceeds from dynamics, describing how cognitive systems operate; EAS proceeds from epistemology, delineating what cognitive systems cannot achieve.

The key divergence: FEP frames cognitive limitations as an **engineering problem** (approximate inference can be progressively improved), while EAS frames them as a **structural theorem** (perfect self-modeling is impossible even with unlimited computation).

As of mid-2026, Friston has not formally addressed Gödelian incompleteness or “perfect self-modeling impossibility” in published work—although the 2025 Fields–Friston paper acknowledged that “a system cannot have a fully precise model of its own processing” [4]. The optimal entry point for dialogue is: **using EAS’s single axiom to provide a first-principles proof of why FEP’s approximate inference can only be approximate.**

## 2 Does FEP Implicitly Contain EAS’s Quotient Cognition Constraint?

### 2.1 FEP’s Core Assumptions

FEP’s argument chain can be summarized in three steps:

1. **Markov blanket demarcation:** Any system that persists over time necessarily possesses a Markov blanket, rendering internal states conditionally independent of external states [1].
2. **Variational free energy upper bound:** Internal states can be interpreted as approximate encodings of posterior distributions; variational free energy is a computable upper bound on surprisal [2].
3. **Self-evidencing:** System dynamics necessarily manifest as free energy minimization—not by choice, but as a consequence of existence.

Crucially, FEP explicitly employs **approximate** Bayesian inference, acknowledging that exact inference is computationally intractable [2]. The 2025 Friston–Sakthivadivel paper further clarifies that FEP is an “as if” description—“using this framework need not presuppose that the system is truly performing inference; it is a useful explanatory fiction” [3].

### 2.2 FEP’s Approximation vs. EAS’s Impossibility

| Dimension                           | FEP                                               | EAS                                  |
|-------------------------------------|---------------------------------------------------|--------------------------------------|
| Nature of cognitive limits          | Computational constraint (tractability)           | Structural impossibility             |
| Terminal state of approx. inference | Can asymptotically approach true posterior        | True posterior is never reached      |
| Self-modeling                       | “Beautiful Loop” enables recursive self-reference | Perfect self-modeling                |
| Source of limitation                | Biological/computational resource limits          | Epistemic structure (incompleteness) |

**Core judgment:** FEP implicitly acknowledges cognitive limitations but frames them as **improvable approximations** rather than **insurmountable theorems**. FEP’s variational free energy  $F \geq -\ln p(o)$  allows equality when the variational density equals the true posterior—meaning FEP’s mathematical structure **preserves the ideal observer as a limit**. EAS’s single axiom asserts that this limit is **epistemically unreachable**—not “hard to reach,” but “structurally impossible.”

### 3 Has Friston (2024–2026) Addressed “Incompleteness”?

#### 3.1 Recent Key Papers

| Paper                                                                                              | Year |
|----------------------------------------------------------------------------------------------------|------|
| Friston, Ramstead & Sakthivadivel, “Generative modelling in non-equilibrium statistical mechanics” | 2025 |
| Fields, Albarracin, Friston et al., “Inner screens and imaginative experience”                     | 2025 |
| Laukkonen, Friston & Chandaria, “A Beautiful Loop”                                                 | 2025 |
| Five Fristonian Formulae (Entropy special issue)                                                   | 2025 |

#### 3.2 Clear Conclusion

As of mid-2026, Friston has not formally addressed Gödelian incompleteness or cognitive boundary problems. Fields–Friston (2025) contains a key statement—“the executive/metacognitive system cannot have a fully-accurate model of the QRFs it interacts with” [4]—but this is presented as a consequence of Ashby’s Required Variety Law, not elevated to an epistemic axiom or insurmountable theorem. The “Beautiful Loop” touches on self-referential structure, but its core thesis is **how self-reference produces consciousness**, not **why self-reference cannot be perfectly closed**.

### 4 Epilexity Five-Layer Resonance vs. FEP Variational Free Energy

#### 4.1 Which Is More Fundamental?

EAS’s Epilexity five-layer resonance validation (internal → cross-domain → predictive → compressive → fractal) is a **validation methodology**—it does not claim to describe how cognition operates, but rather that **any valid epistemic conclusion must pass five-layer resonance to count as validated**. FEP’s variational free energy is a **descriptive principle**—it claims cognitive systems necessarily minimize this quantity.

The two do not compete at the same level:

- **Variational free energy** answers: What do cognitive systems do? (Minimize free energy.)
- **Epilexity five-layer resonance** answers: How do we know an epistemic conclusion is valid? (Must pass five layers; one layer is insufficient.)

#### 4.2 Which Is More Operational?

| Dimension               | Epilexity Five-Layer                       | Variational Free Energy                     |
|-------------------------|--------------------------------------------|---------------------------------------------|
| Mathematical definition | Five-layer validation protocol             | Precisely defined: $F = D_{KL}[q  p] - \ln$ |
| Computability           | Layer-by-layer judgment; no closed formula | Computable via predictive coding etc        |
| Empirical grounding     | Each layer requires independent validation | Extensive predictive coding evidence        |
| Scope                   | Validation of epistemic conclusions        | Any Markov blanket system                   |

**Judgment:** Variational free energy far surpasses in operationality—it has closed mathematical form, computable approximations, and neuroscience implementation paths. Epiplexity five-layer resonance, as a validation framework, provides value in **preventing epistemic overreach**, but lacks the computational tractability of variational free energy.

## 5 Core Divergences

### 5.1 Divergence 1: Cognitive Limits as “Engineering Approximation” vs. “Structural Impossibility”

- **FEP:** Approximate inference exists because exact inference is computationally intractable; in principle, variational density can approach the true posterior arbitrarily closely.
- **EAS:**  $S \subset E \implies R_S \not\cong E$ —even with unlimited computation, perfect self-modeling remains impossible; this is an epistemic-structural necessity.
- **Friston’s possible response:** FEP’s “as if” framework already does not commit to the system possessing the true posterior, so EAS’s constraint is already implicitly acknowledged within FEP.
- **EAS’s rebuttal:** FEP’s mathematical structure preserves  $q = p(\cdot|o)$  as a limit (KL divergence can approach zero), implying the ideal observer is conceivable as an asymptotic limit. EAS explicitly denies this.

### 5.2 Divergence 2: Self-Reference as “Mechanism of Consciousness” vs. “Loop That Cannot Perfectly Close”

- **FEP** (“Beautiful Loop”): Self-referential recursion produces consciousness—“the world model knows itself non-locally” [5].
- **EAS:** Self-reference cannot perfectly close— $R_S \not\cong E$ ; any self-model is necessarily a quotient space, losing information.
- **Friston’s possible response:** Beautiful Loop does not require “perfect” closure—as long as recursion is sufficiently deep, it produces the phenomenon of consciousness. Perfection is not a necessary condition for consciousness.
- **EAS’s rebuttal:** If one admits self-reference is imperfect, then “Beautiful Loop” has seams. EAS theorematizes the size and source of these seams.

### 5.3 Divergence 3: Markov Blanket as “Enabler of Cognition” vs. “Cage of Cognition”

- **FEP:** Markov blanket makes inference possible—without boundaries, there is no inferring agent.
- **EAS:** Markov blanket is precisely the realization of  $S \subset E$ —internal states can only be approximate encodings of external states; the blanket itself is the boundary of the quotient map.
- **Friston’s possible response:** This is exactly FEP’s insight—cognition is achieved through blanket conditional independence, not limited by it.
- **EAS’s rebuttal:** FEP sees only the enabling face, not the limiting face. EAS’s single axiom proves that enabling and limiting are two sides of the same coin.

## 5.4 Divergence 4: Terminal State of Free Energy Minimization

- **FEP**: Non-equilibrium steady state (NESS); variational density approximates the true posterior.
- **EAS**: Even upon reaching NESS, internal states still encode a quotient space of external states— $R_S$  rather than  $E$  itself. Free energy minimization can only shrink the distance between quotient space and original space, not eliminate it.
- **Friston’s possible response**: FEP never claimed internal states fully encode external states—the “as if” framework makes no such commitment.
- **EAS’s**: If FEP does not commit to  $q \rightarrow p$  convergence, then FEP’s description of “self-evidencing” is not about truth but about survival—fundamentally different from EAS’s epistemic concerns.

## 5.5 Divergence 5: Status of Lean4 Formalization

- **EAS**: All theorems T0–T7 compile and pass in Lean4; formal verification ensures unimpeachability.
- **FEP**: FEP lacks equivalent formal verification. Biehl, Pollock & Kanai (2021) proved by counterexample that FEP’s Free Energy Lemma is “wrong in general” [6].
- **Friston’s possible response**: FEP is a physical principle, not a mathematical theorem—analogue to the principle of least action; formal proof is not required.
- **EAS’s rebuttal**: Physical principles can be challenged (FEP has been revised multiple times), while Lean4-verified theorems cannot be overturned. This is a fundamental methodological difference.

# 6 Potential Dialogue Points

## 6.1 Common Language: Both Agree “Cognition Is Bounded”

FEP acknowledges exact inference is infeasible; EAS proves perfect self-modeling is impossible. Both share an intuition: **finite systems cannot fully grasp infinite reality**. The difference lies in the nature of this “finitude”: FEP says “computationally finite,” EAS says “structurally finite.”

## 6.2 EAS Can Provide First-Principles Support for FEP

Dietz’s closure framework [7] independently derives from philosophical first principles a conclusion structurally equivalent to FEP—“free energy is remainder; prediction error minimization is supersession.” EAS’s single axiom arrives at a similar position via a different path: **because  $S \subset E \implies R_S \not\cong E$ , finite systems must minimize free energy**—free energy minimization is the only survival strategy for quotient cognitive systems in NESS. Three frameworks (FEP, Dietz’s closure framework, EAS) converge to the same structure from different directions.

## 6.3 “Incompleteness” as Completion of FEP, Not Negation

EAS is not a replacement for FEP but its epistemic completion. FEP answers “how cognitive systems operate”; EAS answers “why cognitive systems cannot operate better.” Together, they form a complete picture.

## 7 Conclusion

1. **FEP has not yet addressed the “incompleteness” problem.** The 2025 Fields–Friston paper touches on this topic but does not theorem-ize it. This is the core gap EAS can fill.
2. **FEP does not assume an ideal observer, but preserves the ideal observer as a mathematical limit.** EAS proves this limit is unreachable—not due to insufficient computational resources, but due to epistemic structure.
3. **Epiplexity five-layer resonance and variational free energy operate at different levels.** The former is a validation methodology; the latter is a descriptive principle. The former is more rigorous but harder to operationalize; the latter is more computable but lacks epistemic foundations.
4. **Optimal dialogue entry point:** Using EAS’s single axiom to provide a first-principles proof for FEP’s claim that “approximate inference can only be approximate”—this is not challenging FEP, but completing its philosophical foundations.



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